Channel1

STEP1 STEP

DIR1 DIR

File: ChannelTop.kicad_sch

Channel2

STEP2 STEP

DIR2 DIR

File: ChannelTop.kicad_sch

Channel3

STEP3 STEP

DIR3 DIR

File: ChannelTop.kicad_sch

Channel4

STEP4 STEP

DIR4 DIR

File: ChannelTop.kicad_sch

Channel5

STEP5 STEP

DIR5 DIR

File: ChannelTop.kicad_sch

Channel6

STEP6 STEP

DIR6 DIR

File: ChannelTop.kicad_sch

Channel7

STEP7 STEP

DIR7 DIR

File: ChannelBottom.kicad_sch

Channel8

STEP8 STEP

DIR8 DIR

File: ChannelBottom.kicad_sch

Channel9

STEP9 STEP

DIR9 DIR

File: ChannelBottom.kicad_sch

Channel10

STEP10 STEP

DIR10 DIR

File: ChannelBottom.kicad_sch

Channel11

STEP11 STEP

DIR11 DIR

File: ChannelBottom.kicad_sch

Channel12

STEP12 STEP

DIR12 DIR

File: ChannelBottom.kicad_sch

Control

File: Control.kicad_sch

Power

File: Power.kicad_sch

Sheet: /
File: 12StepperPCB.kicad_sch

**Title: 12-Channel Stepper Driver Carrier**

| Size: A2 | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 1/15 |

VMOT: 30V max, 12A bus (1A/ch target). Dual entry, second terminal unfused —
for low-resistance current injection only, not a redundant protection path.
A4988 max 35V; TMC2209 max 28V operating (29V abs max). Above 28V: A4988-only.
Fuse footprint left TBD — confirm blade-fuse holder or polyfuse P/N before layout.

J51




VDD_5V_IN

J52




VMOT_IN_A

J53




VMOT_IN_B



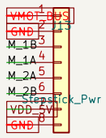


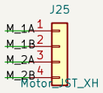





| | | |
|---|---|---|
| Sheet: /Power/ | | |
| File: Power.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 2/15 |

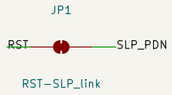


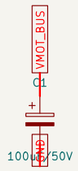



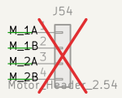





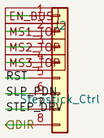


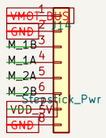


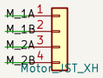




| | | |
|---|---|---|
| Sheet: /Channel1/ | | |
| File: ChannelTop.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 3/15 |

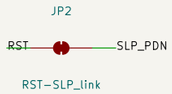


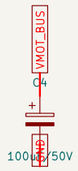



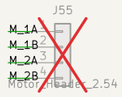





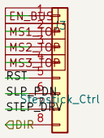


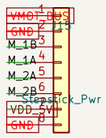


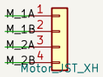




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|---|---|---|
| Sheet: /Channel2/ | | |
| File: ChannelTop.kicad_sch | | |
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| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 4/15 |

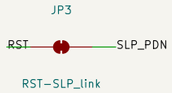


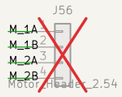






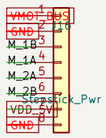


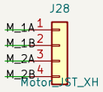





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|---|---|---|
| Sheet: /Channel3/ | | |
| File: ChannelTop.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 5/15 |

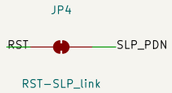


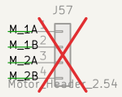




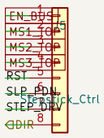


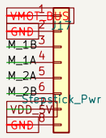


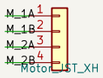





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|---|---|---|
| Sheet: /Channel4/ | | |
| File: ChannelTop.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 6/15 |

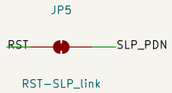


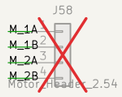




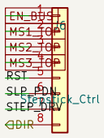


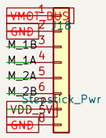


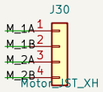





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|---|---|---|
| Sheet: /Channel5/ | | |
| File: ChannelTop.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 7/15 |

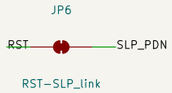


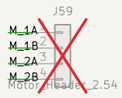






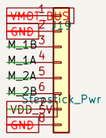


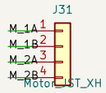





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|---|---|---|
| Sheet: /Channel6/ | | |
| File: ChannelTop.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 8/15 |

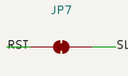


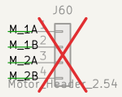




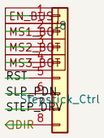


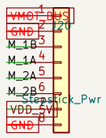


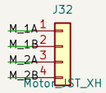





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|---|---|---|
| Sheet: /Channel7/<br>File: ChannelBottom.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 9/15 |



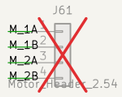




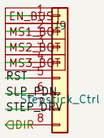


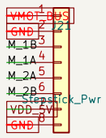


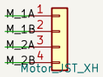





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|---|---|---|
| Sheet: /Channel8/ | | |
| File: ChannelBottom.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 10/15 |

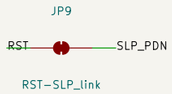


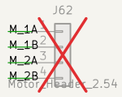




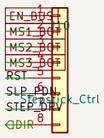


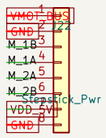


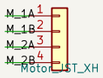





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|---|---|---|
| Sheet: /Channel9/ | | |
| File: ChannelBottom.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 11/15 |

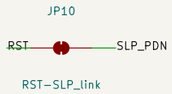


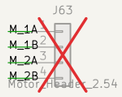








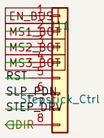


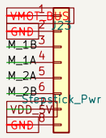


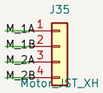

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File: ChannelBottom.kicad_sch

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| KiCad E.D.A. 10.0.5 | | Id: 12/15 |

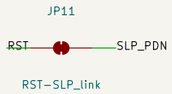


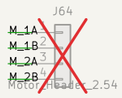




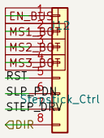


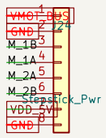


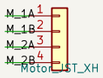





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| KiCad E.D.A. 10.0.5 | | Id: 13/15 |

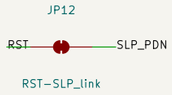


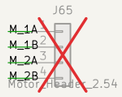










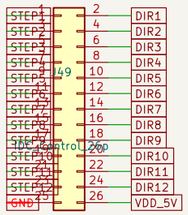


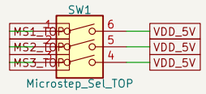

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| KiCad E.D.A. 10.0.5 | | Id: 14/15 |

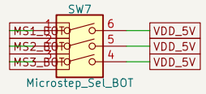
EN header: center=EN_BUS, pin1=GND (jumper 1–2 = always enabled),
pin3=EN_MCU breakout for MCU–driven global enable (external jumper to center).




2–SWITCH VARIANT: SW1 sets MS1–MS3 for channels 1–6 (top row),
SW7 for channels 7–12 (bottom row). All channels in a row share microstepping.




el): odd pins 1,3..23=STEP1–12, even pins 2,4..24=DIR1–12, 25=GND, 26=VDD (5V sense/supply). Each channel's STEP/DIR pair shares one connector column and adjacent ribbon wires.

| | | |
|---|---|---|
| Sheet: /Control/ | | |
| File: Control.kicad_sch | | |
| **Title:** | | |
| Size: A3 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.5 | | Id: 15/15 |